EXPERIMENT 6

6.1 PROGRAMS

6.1.1 Incremented Date

Ans : **ALGORITHM:**

Step 1: Start
Step 2: Read d, m, y
Step 3: Set flag = 1

Step 4: If y < 0 OR m < 0 OR m > 12
→ Set flag = 0

Step 5: Determine maxdays:

- If m in (1,3,5,7,8,10,12)
  maxdays = 31
- Else if m in (4,6,9,11)
  maxdays = 30
- Else (February case)
  If y % 2 == 0
  maxdays = 29
  Else
  maxdays = 28

Step 6: If d > maxdays
→ Set flag = 0

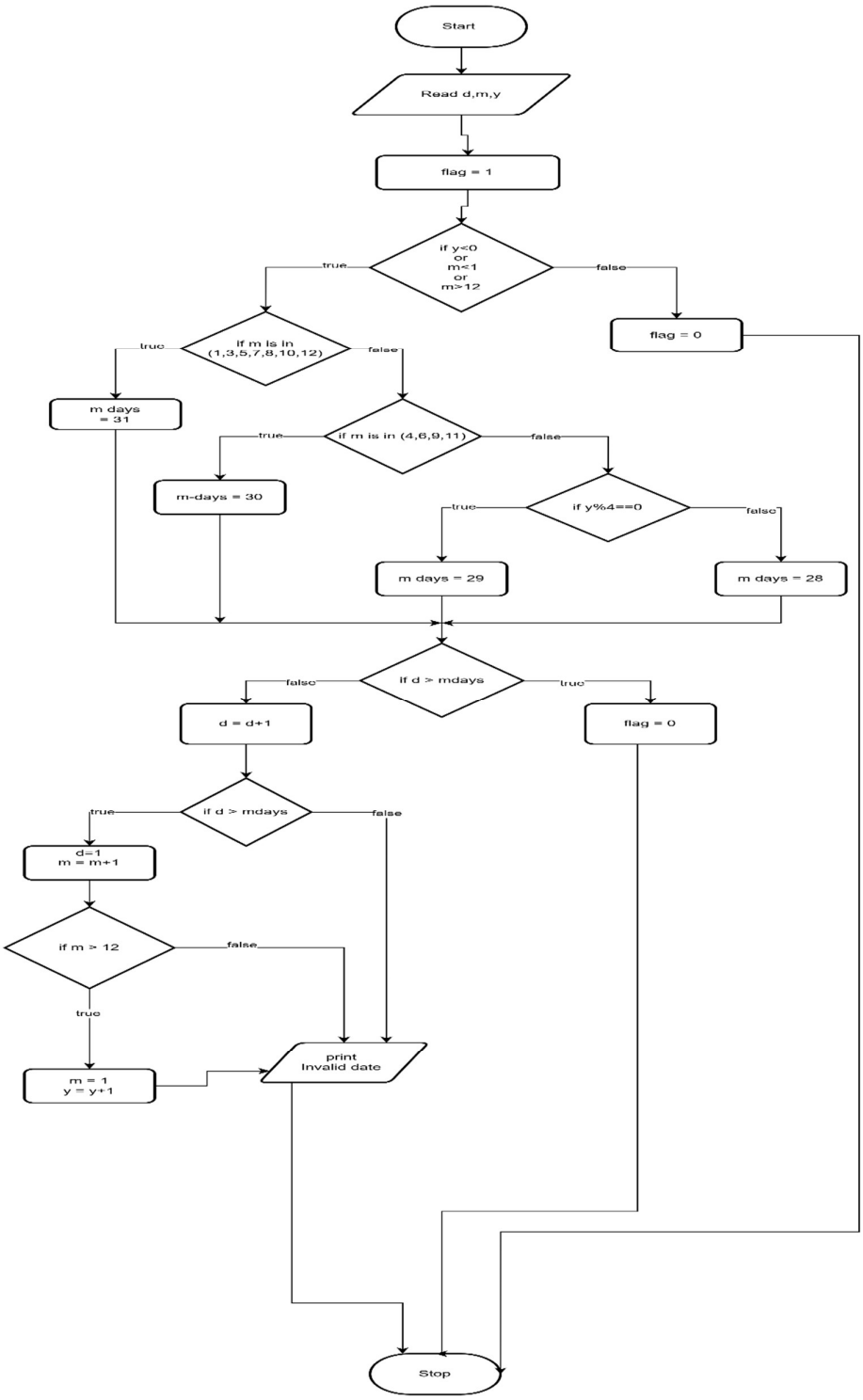
Step 7: Increment date:

- d = d + 1
- If d > maxdays
  d = 1
  m = m + 1

Step 8: If m > 12
m = 1
y = y + 1

Step 9: If flag == 1
Print date in format DD-MM-YYYY
Else
Print "Invalid Date"

Step 10: Stop


Start
Read d,m,y
flag = 1
if y<0
or
m<1
or
m>12
true
false
flag = 0
if m is in
(1,3,5,7,8,10,12)
true
false
m days
= 31
if m is in (4,6,9,11)
true
false
m-days = 30
if y%4==0
true
false
m days = 29
m days = 28
if d > mdays
false
true
d = d+1
flag = 0
if d > mdays
true
false
d=1
m = m+1
if m > 12
false
true
m = 1
y = y+1
print
Invalid date
Stop

FLOWCHART:

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6.1.1. Incremented Date 39:54

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. **Valid Month**: 1 to 12
2. **Valid Day**: Depends on the month and year
   - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
   - April (4), June (6), September (9), November (11): 1 to 30 days
   - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. **Valid Year**: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Sample Test Cases +

Explorer

nextDate.py Submit

```python
# Type Content here...
d = int(input())
m = int(input())
y = int(input())

flag = 1

if y<0 or m<0 or m>12:
    flag = 0
if m in (1,3,5,7,8,10,12):
    maxdays = 31
elif m in (4,6,9,11):
    maxdays = 30
else:
    if y % 2 == 0:
        maxdays = 29
    else:
        maxdays = 28

if d > maxdays:
    flag = 0

d = d + 1
if d>maxdays:
    d = 1
```

Debugger

Terminal Test cases

Prev Reset Submit Next

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Sample Test Cases +

Explorer

nextDate.py Submit

```
    maxdays = 31
elif m in (4,6,9,11):
    maxdays = 30
else:
    if y % 2 == 0:
        maxdays = 29
    else:
        maxdays = 28

if d > maxdays:
    flag = 0

d = d + 1
if d>maxdays:
    d = 1
    m = m+1
if m > 12 :
    m = 1
    y =y+1

if flag == 1:
    print(f"{d:02d}-{m:02d}-{y}")
else:
    print("Invalid Date")
```

Debugger

Terminal Test cases

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- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Sample Test Cases +

Explorer

nextDate.py Submit

```
    d = 1
    m = m+1
if m > 12 :
    m = 1
    y =y+1

if flag == 1:
    print(f"{d:02d}-{m:02d}-{y}")
else:
    print("Invalid Date")
```

Debugger

Average time	Maximum time
0.009 s	0.013 s
8.60 ms	13.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 13 ms Debug

Expected output	Actual output
15	15
3	3
2024	2024
16-03-2024	16-03-2024

Test case 2 10 ms

Terminal Test cases

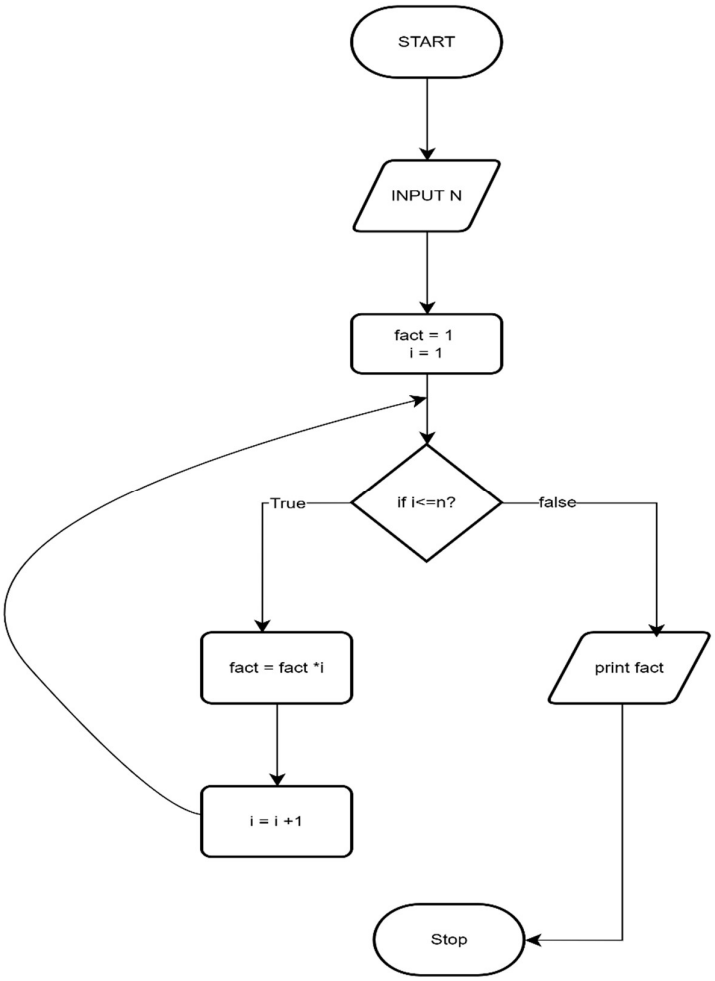
Prev Reset Submit Next

6.1.2//Factorial of a number

Ans: ALGORITHM:

1. **Start**
2. Input number n
3. Set fact = 1
4. Set loop variable i = 1
5. Check if i ≤ N
    - If **Yes** → go to step 6
    - If **No** → go to step 8
6. Compute fact = fact × i
7. Increment i = i + 1 and go back to step 5
8. Print fact
9. **Stop**

FLOWCHART:


START
INPUT N
fact = 1
i = 1
if i<=n?
True
false
fact = fact *i
print fact
i = i +1
Stop

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6.1.2. Factorial of a Number

01:51

Write a Python program to calculate the factorial of a number n using loops.

Input Format:

- A single line containing an integer n.

Output Format:

- Print the factorial of the given integer n.

Sample Test Cases +

factorialN... Submit

```
#Write your code here
n = int(input())

fact = 1
for i in range(1,n+1):
    fact = fact*i

print(fact)
```

Average time	Maximum time
0.003 s	0.005 s
3.25 ms	5.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 5 ms Debug

Expected output	Actual output
10	10
3628800	3628800

Test case 2 2 ms

Terminal Test cases

Prev Reset Submit Next